Gurucharan Vemuru

Milwaukee, Wisconsin, United States | gurucharanvemuru@gmail.com | (414) 629-1607 | github.com/gvemuru1 |
linkedin.com/in/gurucharanvem

Professional Summary

AI Engineer and Software Engineer with 3+ years of experience across backend systems, full-stack products, and cross-platform mobile development. Builds AI-driven applications by combining retrieval and LLM workflows with production software engineering practices. Background spans distributed systems, APIs, cloud-native services, accessible web experiences, and React Native applications, with a focus on reliable products that connect engineering decisions to user and business outcomes.

Skills

AI / LLM Engineering: RAG, Agentic RAG, LangChain, LangGraph, Embeddings, Prompt Grounding, Query Expansion, Contextual Compression, LangSmith

Retrieval & Vector Data: ChromaDB, PostgreSQL/pgvector, Semantic Search, BM25, Hybrid Search, Reciprocal Rank Fusion (RRF), Metadata Filtering, HNSW

Software & Native: Python, Go, JavaScript, TypeScript, React, Next.js, React Native/Expo, Node.js, Express.js, FastAPI, REST, GraphQL, gRPC

Data, Cloud & Systems: PostgreSQL, MongoDB, Redis, TensorFlow/Keras, AWS, Docker, Kubernetes, Kafka, RabbitMQ, CI/CD, Jaeger, OpenTelemetry, OAuth 2.0, JWT, RBAC, WCAG/ADA

Work Experience

Software Engineer Jul 2025 - Aug 2026

Zably AI

- Designed Node.js and FastAPI microservices with gRPC and RabbitMQ to support low-latency service communication and asynchronous product workflows.
- Built an on-demand marketplace MVP across Next.js and React Native/Expo, taking customer flows from Figma prototypes through cross-platform implementation and App Store compliance.
- Managed Kubernetes deployments with Tilt and Jaeger tracing, improving developer feedback loops and making cross-service latency easier to diagnose.
- Developed Google Maps and OSRM routing services for real-time proximity matching and location search, supporting core marketplace discovery experiences.

Web Accessibility Developer Sep 2024 - May 2025

UWM Accessibility Resource Center

- Evaluated React applications and university web platforms against WCAG 2.1 AA, ARIA, and ADA requirements, improving access to digital services for users of assistive technologies.
- Refactored HTML and frontend components to resolve keyboard, contrast, and screen-reader issues while supporting institutional accessibility compliance.

Software Engineer Apr 2021 - Jun 2023

Stack Apps

- Integrated Kafka into financial workflows using at-least-once delivery and idempotent consumers, protecting payment systems from duplicate transaction processing.
- Built Go microservices with OAuth 2.0 and RBAC controls, contributing to PCI-DSS initiatives and reducing security incidents by 35%.
- Developed PostgreSQL procedures and triggers supporting 5,000+ monthly financial transactions, reducing manual operational work by 30% and lowering error rates below 2%.

Projects

Production RAG & Agentic AI Retrieval System

- Implemented an end-to-end RAG pipeline for document ingestion, recursive chunking, embeddings, vector persistence, similarity retrieval, source-aware context formatting, and grounded LLM responses.
- Built ChromaDB and PostgreSQL/pgvector retrieval paths; combined semantic vector search with BM25 keyword retrieval and Reciprocal Rank Fusion for stronger exact-term and semantic matching.
- Structured multi-step retrieval and generation workflows with LangGraph and FastAPI, keeping model, retriever, vector-store, and API components modular for AI-driven application development.
- Applied metadata filtering, similarity scoring, source attribution, and retrieval-quality techniques to improve answer grounding and make RAG behavior easier to evaluate and debug.

Doodle Recognition Web Application

- Trained a 7-layer CNN on 414,000+ vector images, converted it to TensorFlow Lite, and integrated it with a FastAPI image-processing backend and interactive browser canvas.

Education

University of Wisconsin-Milwaukee

M.S., Information Technology — *May 2025*

B.E., Computer Science — *May 2022*